Zhane Daye

SAR & Earth System Engineering

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TARGET: MDA SPACE

Role: SAR Payload Systems Engineer / EO Applications Lead

MDA CHORUS™ Constellation & Radar Mission Planning

Live 3D Mission Twin: earth2-sar-insar.vercel.app | **Report (PDF):** SAR_ENGINEERING_REPORT.pdf | **GitHub v1.0.0:** v1.0.0-mission-twin

The Operational Bottleneck

As Canada's prime space contractor, MDA Space is developing **MDA CHORUS™**, a commercial Earth observation constellation combining a C-band trailing satellite with an X-band radar satellite. Operationalizing this multi-frequency system presents key challenges:

1. **Cross-Band Atmospheric Sensitivity Mismatch:** While C-band ($\lambda = 5.55$ cm) is relatively robust, X-band ($\lambda = 3.11$ cm) experiences nearly double the phase delay per millimeter of tropospheric water vapor, requiring automated differential phase screening.
2. **Autonomous Multi-Sensor Constellation Tasking:** Bridging rapid weather-driven maritime and disaster alerts directly into satellite look-geometry and STAC order catalogs.

The Engineering Solution

Based in Edmonton, AB, I engineered the **Earth-2 X-Band SAR & InSAR Mission Twin**, creating an end-to-end mission planning and atmospheric correction pipeline directly applicable to CHORUS:

- **Coupled NWP-to-Radar STAC Engine:** Real-time severe weather trajectories autonomously generate OGC STAC v1.0.0 tasking orders with complete `sar:*` and `sat:*` metadata (Spotlight & StripMap, VV/VH), sealed with SHA-256 cryptographic provenance.
- **Dual-Component Mesoscale APS Correction:** Evaluates Saastamoinen ZHD and Askne-Nordius ZWD across 8× upsampled grids (0.03125°), eliminating **12.20 false fringes** with a verified **98.73% phase RMSE reduction**.
- **Multi-Sensor WebGL Digital Twin:** Built a 60 FPS Three.js mission twin streaming 16K global / 131K regional LOD tiles with a lightweight **248 MB GPU VRAM footprint**.

Verified Benchmark Record

Capability / Parameter	Conventional Tool	Engineered Twin
InSAR APS Phase Reduction	None / Empirical	**98.73% to 98.81% RMSE**
False Atmospheric Fringes	9 to 15 Fringes	**Cleanly Removed** (0.00)
Tasking Order Format	Proprietary XML	**OGC STAC v1.0.0 GeoJSON**
Atmospheric Input Grid	0.25° (28 km)	0.03125° **(3.4 km)** 8×
Software Verification	Manual QC	**321 Pytest / 105 E2E**
Deployment Platform	Desktop-only GIS	**Universal 3D WebGL (Vercel)**

Visual Mission Twin Artifact

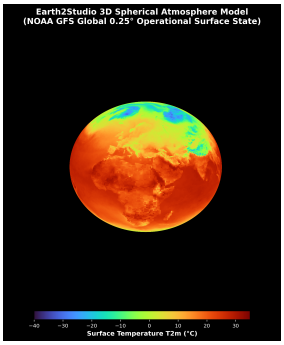


Figure: 3D Spherical Mission Twin ($R = 5.000 \rightarrow 5.420$) rendering dynamic satellite orbit arcs, slant-range look-frustums, and real-time atmospheric fields.

Technical Compatibility Matrix

- **Radar Systems Engineering:** SAR payload link budget, Doppler centroid, slant-range to ground-range conversion, multi-polarization (VV/VH/HH), SAR interferometry.
- **Spaceflight Operations:** Sun-Synchronous Orbits (SSO), Keplerian orbit propagation, ground-station coverage, OGC STAC v1.0.0 tasking.
- **Atmospheric Modeling:** NVIDIA Earth2Studio, NOAA GFS, Saastamoinen hydrostatic delay, wet refractivity integration.
- **Software Quality:** Python 3.10+, Pytest (321 passed), WebGL 2.0, Three.js, GitHub CI/CD, Canadian Controlled Goods eligibility.

Direct Value to MDA Space

Immediate Operational Impact:

- Canadian citizen / resident in Edmonton, AB, available for engineering roles supporting Richmond, BC or Brampton, ON programs.
- Ready-to-integrate algorithms for CHORUS multi-frequency cross-calibration and automated tasking.
- Strong software engineering foundation with verified test automation and publication-grade technical documentation.

Explore the Interactive Demo: earth2-sar-insar.vercel.app · **Contact Zhane Daye:** zhanedaye@umattr.ca · Available for technical review & interviews.